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Occupational and socioeconomic predictors of myocardial infarction and coronary heart disease: a machine learning analysis

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Abstract

Background Cardiovascular disease (CVD) remains a leading cause of morbidity and mortality worldwide. Although traditional cardiovascular risk models primarily rely on biomedical factors, socioeconomic and occupational characteristics are increasingly recognized as important correlates of cardiovascular health. However, applying machine learning to population-based survey data raises methodological concerns, particularly reverse causation and post-diagnosis information leakage.

Methods We conducted a cross-sectional analysis using data from the 2023 Behavioral Risk Factor Surveillance System (BRFSS). The analytic objective was classification of prevalent myocardial infarction (MI) or coronary heart disease (CHD) rather than prospective risk prediction. Four supervised machine learning algorithms (logistic regression, decision tree, random forest, and gradient boosting) were evaluated. To address potential label leakage, we implemented two model variants: an inclusive model incorporating all available predictors (Model A) and a model excluding post-diagnosis proxy variables such as medication use, functional limitations, and disability indicators (Model B). Model performance was assessed using receiver operating characteristic area under the curve (ROC-AUC), precision–recall area under the curve (PR-AUC), precision, recall, and F1 score.

Results Gradient boosting demonstrated the strongest discriminative performance among the evaluated models. In the inclusive setting, the model achieved a ROC-AUC of 0.867 and a PR-AUC of 0.389, with a best F1 score of 0.433. After removal of post-diagnosis variables, performance remained robust (ROC-AUC = 0.858; PR-AUC = 0.372; F1 = 0.418), suggesting that predictive capacity was not driven solely by downstream disease indicators. Feature

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importance analyses showed that socioeconomic and employment-related variables remained prominent predictors in Model B alongside established clinical risk factors.

Conclusions Machine learning models can effectively classify prevalent MI and CHD using large-scale survey data even after explicit mitigation of post-diagnosis information leakage. Socioeconomic and occupational characteristics appear to function primarily as contextual correlates rather than causal determinants of CVD. These findings highlight the value of interpretable machine learning approaches for the population-level classification of prevalent cardiovascular disease while underscoring the limitations inherent to cross-sectional data.

Keywords Cardiovascular disease, Machine learning, Social determinants of health, Occupational factors, Public health

Introduction

Cardiovascular disease (CVD) remains one of the leading causes of death in the United States [1]. In 2021, it caused 928,741 deaths, accounting for 20% of all deaths in the country [2]. During the same year, the global death toll from CVD was approximately 19.1 million, representing 32% of all deaths worldwide [2]. CVD, which remains the primary cause of mortality both in the United States and globally, also imposes a substantial socioeconomic burden. From 2020 to 2021, the total direct and indirect costs associated with CVD in the United States were estimated at $417.9 billion per year [2]. Understanding the factors associated with existing CVD is important for identifying high-burden populations and informing population-level understanding of disease burden [3].

Traditional cardiovascular risk assessment tools primarily rely on physiological and laboratory measurements. While these approaches have demonstrated clinical utility, they may not fully capture the broader social and environmental contexts that shape cardiovascular health. Increasing evidence suggests that socioeconomic conditions, employment status, access to care, and lifestyle environments contribute meaningfully to CVD patterns at the population level [4–8].

Recent advances in machine learning have enabled the analysis of complex, high-dimensional health data and have increasingly been applied to classification of cardiovascular disease. Several studies have reported that machine learning models can achieve improved discrimination compared with traditional risk scores when applied to large clinical or population-based datasets [9–11]. In addition to conventional biomedical predictors, recent models have incorporated social determinants of health such as socioeconomic status, employment conditions, and access to healthcare [4–6]. These approaches suggest that socioeconomic and environmental factors may provide additional information beyond traditional physiological risk indicators. However, applying machine learning to cross-sectional population survey data introduces important methodological challenges. Reverse causation and post-diagnosis information leakage may arise when variables reflect downstream consequences of disease, such as treatment patterns, disability, or perceived health status [12, 13].

Accordingly, the objective of this study was not prospective risk prediction, but rather classification of prevalent myocardial infarction (MI) and coronary heart disease (CHD) using cross-sectional survey data. We aimed to (1) develop and compare multiple supervised ML algorithms; (2) explicitly evaluate the impact of post-diagnosis proxy variables through parallel model specifications; and (3) assess the contribution of socioeconomic and employment-related variables within a framework designed to mitigate label leakage.

Methods

Study design and analytic framework

This study employed a cross-sectional design using the 2023 Behavioral Risk Factor Surveillance System (BRFSS). The scientific task was classification of prevalent MI or CHD based on self-reported survey responses. Because the outcome reflects existing disease status at the time of survey, the models classify prevalent CVD status based on self-reported data rather than estimate future cardiovascular risk.

Two model variants were constructed to address reverse causation. Model A (Inclusive) included all eligible predictors after removal of variables directly encoding MI/CHD diagnosis. Model B (post-diagnosis-excluded model) excluded a priori defined post-diagnosis proxy variables (e.g., medication use, functional limitations, and disability indicators) to reduce information leakage and evaluate the robustness of model performance. The complete list of excluded proxy variables is shown in Supplementary Table (1) Model B was implemented as a sensitivity analysis to assess whether model performance was driven by post-diagnosis information rather than antecedent health characteristics. The comparison between Model A and Model B therefore serves as an internal validity check for potential target leakage rather than a comparison of alternative model specifications. The detailed model specifications and sensitivity analysis framework are summarized in Supplementary Table (2) The overall analytic pipeline is summarized in Fig. 1.

Dataset used

BRFSS is a nationally representative, annual telephone survey conducted across the United States. The 2023

dataset includes responses from 433,323 individuals and contains approximately 350 health-related, behavioral, and sociodemographic variables.

Outcome definition

The primary outcome was self-reported history of MI or CHD (_michd) in the BRFSS survey. Participants were classified as positive if they answered that a healthcare professional had ever informed them that they had MI or CHD. Because this variable represents prior physician-diagnosed disease reported at the time of survey, it reflects prevalent CVD rather than incident events. Accordingly, the present analysis should be interpreted as classification of existing disease status and not prospective cardiovascular risk prediction.

Data cleaning and handling

Non-health administrative variables were excluded. Variables with entirely missing values were removed. All BRFSS non-substantive response codes (e.g., "Don't know/Not sure," "Refused," and survey-specific missing codes) were recoded as missing according to the BRFSS codebook.

Variables directly overlapping with the outcome (e.g., explicit MI/CHD diagnosis indicators) were removed a priori to prevent trivial label leakage. The full list of variables excluded prior to model training is provided in Supplementary Table 3. These exclusions were determined before model training based on clinical plausibility to prevent the model from learning consequences of prior diagnosis rather than correlates of disease status. After these exclusions, 348 candidate predictors were retained for Model A. Model B further excluded the predefined post-diagnosis proxy variable set defined a priori based on clinical plausibility.

Post-diagnosis proxy variables were grouped into three categories: (1) medication and treatment indicators, (2) current health status measures, and (3) functional limitation or disability indicators. These variables included cholesterol-lowering medication use, blood-pressure medication use, self-rated general health, and functional limitation indicators such as difficulty walking or climbing stairs. These variables were excluded from Model B before any preprocessing, tuning, or model fitting and were never provided to the learning algorithms.

Preprocessing pipeline

Given the mixed data types in the BRFSS dataset, the data were first divided into training and test sets using stratified sampling based on the outcome variable (70:30 split, random seed = 42) to preserve the prevalence of MI/CHD. Continuous variables were imputed using the median value, whereas categorical variables (both nominal and ordinal) were imputed using the most frequent category (mode). Nominal variables were encoded using one-hot encoding, and ordinal variables were encoded using ordinal encoding to preserve the inherent ordering of response categories as defined in the original BRFSS survey.

All preprocessing procedures were fitted exclusively on the training dataset and subsequently applied unchanged to the test dataset to prevent information leakage. Specifically, imputation values and encoding mappings were derived solely from the training data and then applied to the test data without modification. No information from the test dataset was used during preprocessing, model training, hyperparameter tuning, or threshold selection.

To evaluate the stability and generalizability of model performance, five-fold stratified cross-validation was conducted within the training dataset. Within each fold, preprocessing steps including imputation and encoding were fitted only on the training portion and then applied to the corresponding validation portion. The independent test set was reserved exclusively for the final performance evaluation and was not used during model selection or threshold determination.

Model development and threshold selection were performed within the training dataset using five-fold stratified cross-validation. Model performance during cross-validation was summarized as mean ± standard deviation for balanced accuracy, ROC-AUC, precision–recall area under the curve (PR-AUC), and F1 score. The independent test dataset was used once for the final hold-out evaluation.

Model development

Four supervised machine learning algorithms were evaluated: logistic regression, decision tree, random forest, and gradient boosting. Model hyperparameters were prespecified based on standard practice and applied consistently across analyses, rather than optimized through an exhaustive grid-search procedure. Classification thresholds were not determined uniformly across all models. For random forest and gradient boosting, thresholds were selected to satisfy predefined performance criteria (sensitivity ≥ 0.80 and specificity ≥ 0.70) when possible. For decision tree and logistic regression models, a default threshold of 0.5 was used. The full set of hyperparameter values used for each model is provided in Supplementary Table 4.

Because the outcome prevalence was relatively low (positive rate $\approx 8.5\%$), class imbalance was addressed using inverse-frequency weighting. Class weights were applied to logistic regression, decision tree, and random forest models, whereas sample weights were used for gradient boosting. All weights were calculated using the training dataset only. All analyses were conducted in Python (version 3.12.7) using scikit-learn (version 1.5.1). A fixed random seed of 42 was used for all data splits

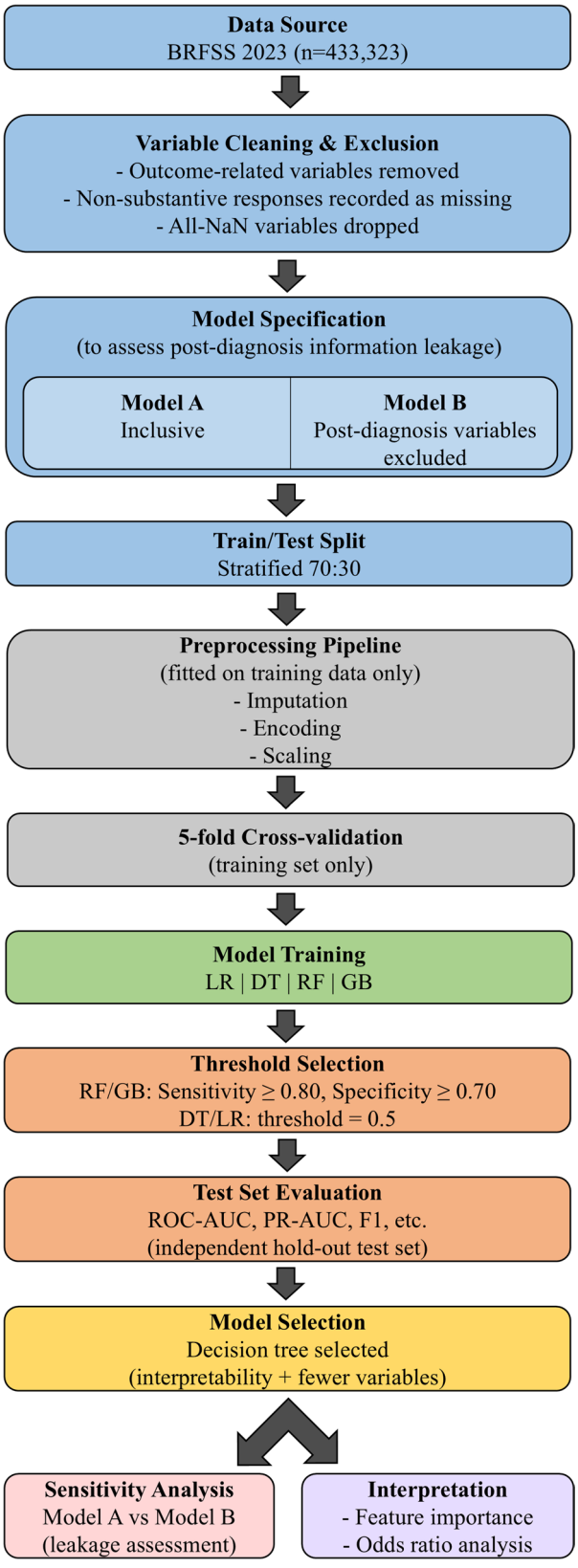


Fig. 1 Schematic overview of the analytic pipeline. The workflow includes data cleaning and variable exclusion, model specification (Model A: inclusive; Model B: post-diagnosis variables excluded), stratified train/test splitting (70:30), preprocessing (imputation, encoding, and scaling), five-fold cross-validation, model training (logistic regression, decision tree, random forest, and gradient boosting), threshold selection, and final evaluation on an independent hold-out test set. All preprocessing, model development, and threshold selection steps were performed exclusively within the training dataset to prevent information leakage. Sensitivity analyses comparing Model A and Model B were conducted to assess the impact of post-diagnosis variables. Final model interpretation was performed using feature importance and odds ratio analyses

and model training procedures to ensure reproducibility. Although all four algorithms were evaluated, the decision tree model was selected for further analysis based on its ability to meet predefined performance criteria (sensitivity ≥ 0.80 and specificity ≥ 0.70) while maintaining interpretability and using a limited number of predictors. The variables included in the odds-ratio analyses were derived from the decision tree models that met the predefined performance criteria (sensitivity ≥ 0.80 and specificity ≥ 0.70). Variables identified from both decision tree specifications were combined to form the final set of predictors. To account for the complex survey design of BRFSS, additional weighted analyses were conducted as a sensitivity analysis. Specifically, multivariable logistic regression models were fitted using BRFSS sampling weights (_cllcpwt) as a sensitivity analysis to assess the robustness of associations under the complex survey design. Machine learning models were not weighted, as standard implementations do not readily accommodate complex design features such as stratification, clustering, and weighting within a unified framework.

Threshold selection and evaluation metrics

Given the class imbalance, model performance was evaluated using multiple metrics rather than accuracy alone. The following metrics were used: receiver operating characteristic area under the curve (ROC-AUC), PR-AUC, precision, recall (sensitivity), and F1 score. All classification metrics derived from predicted labels (including precision and F1 score) were calculated with MI/CHD defined as the positive class (class 1). In addition, both weighted and unweighted prevalence estimates were calculated to assess the impact of sampling weights on the distribution of the outcome.

Classification thresholds were not determined uniformly across all models. For random forest and gradient boosting, thresholds were selected to satisfy predefined performance criteria (sensitivity ≥ 0.80 and specificity ≥ 0.70) when possible. For decision tree and logistic regression models, a default threshold of 0.5 was used. The selected threshold was then applied unchanged

to the independent test dataset for final performance evaluation.

In addition, probabilistic calibration of the models was assessed using calibration curves, the Brier score, and calibration intercept and slope. Calibration intercept and slope were estimated using a logistic recalibration model, in which the observed outcome was regressed on the logit of predicted probabilities in the independent test dataset. Calibration analyses were conducted for the final models under both Model A and Model B specifications. The independent test dataset was used only once for the final performance evaluation after model selection and threshold determination.

In addition to machine learning analyses, odds ratios (ORs) were estimated using variable-specific logistic regression models. For each predictor, a separate (univariable) logistic regression model was fitted with MI/CHD (_MICHD) as the outcome (1 = yes, 0 = no).

Binary variables were coded as 1 (presence) and 0 (absence), and responses coded as "don't know" or "refused" were treated as missing. Categorical variables were modeled using indicator variables with predefined reference categories.

Missing data were handled using complete-case analysis, whereby observations with missing values in either the outcome or the predictor were excluded prior to model fitting.

Analysis of key variables

To enhance interpretability, analyses focused primarily on decision tree–based impurity importance, which allows transparent rule-based interpretation of predictors. Feature importance or model coefficients from the other algorithms were also reviewed descriptively to assess the consistency of key predictors across models. This approach enabled differentiation between variables likely reflecting downstream consequences of disease and those representing broader demographic, socioeconomic, and comorbidity-related correlates of prevalent MI/CHD.

Variable importance analysis

Variable importance for the decision tree models was assessed using both impurity-based feature importance (Gini importance) and permutation-based feature importance.

Gini importance, implemented in the scikit-learn DecisionTreeClassifier, quantifies the total reduction in Gini impurity contributed by each feature across all decision splits in the final trained model. However, this measure is known to be sensitive to data structure and model-specific splitting behavior.

To address these limitations, permutation importance was additionally computed on the independent test dataset (held-out test set). For each feature, values were randomly permuted multiple times ($n = 20$), and the resulting mean decrease in model performance (measured by ROC-AUC) was used to quantify importance. This approach provides a model-agnostic and more robust assessment of feature importance.

Feature importance rankings were interpreted by comparing both Gini-based and permutation-based measures, and consistency across these methods was considered when identifying key predictors. All importance measures were interpreted as model-specific and exploratory rather than causal estimates.

Results

Study population

The final analytic dataset included 433,323 respondents from the 2023 BRFSS. The prevalence of self-reported MI or CHD was 8.47%, reflecting substantial class imbalance. Demographic, socioeconomic, and health-related characteristics were consistent with the nationally representative design of the survey.

Model performance

A comprehensive comparison of all four algorithms is presented in Table 1. Across all evaluated machine learning algorithms, gradient boosting demonstrated the strongest discriminative performance (ROC-AUC = 0.864; PR-AUC = 0.381). Although gradient boosting achieved the highest overall performance, the decision tree model was selected for subsequent analyses because it satisfied predefined performance criteria while using a substantially smaller number of predictors, thereby improving interpretability. Additional analyses were conducted using decision tree models under two specifications (Model A and Model B) to evaluate the impact of excluding post-diagnosis proxy variables. The detailed performance of the decision tree models is presented in Table 2. In Model A (inclusive), the gradient boosting model achieved a ROC-AUC of 0.867 and a precision–recall AUC (PR-AUC) of 0.389, with a best F1 score of 0.433 at the optimal classification threshold. In Model B, which excluded post-diagnosis proxy variables, performance remained robust (ROC-AUC = 0.858; PR-AUC = 0.372; F1 score = 0.418). Although modest reductions in performance were observed, overall discrimination was preserved, suggesting that predictive capacity may not be entirely explained by downstream disease indicators. Decision tree and random forest models exhibited similar patterns, with consistently higher performance in Model A compared with Model B, but without catastrophic degradation following removal of post-diagnosis variables.

Five-fold stratified cross-validation demonstrated stable performance for both decision tree variants

Table 1 Predictive performance of machine learning models applied to the BRFSS 2023 dataset

Model specification	N_features	Threshold	ROC-AUC	AUPCR	Accuracy	Balanced accuracy	Sensitivity	Brier score	Specificity	Precision	F1 score	CV ROC-AUC (mean ± SD)	CV AUPRC (mean ± SD)
Decision tree	11	0.50	0.830	0.305	0.719	0.760	0.809	0.174	0.710	0.205	0.327	0.832 ± 0.001	0.308 ± 0.003
Random forest	216	0.538	0.838	0.332	0.731	0.762	0.800	0.178	0.725	0.211	0.334	0.839 ± 0.002	0.330 ± 0.003
Logistic regression	338	0.50	0.860	0.375	0.769	0.785	0.805	0.156	0.765	0.240	0.370	0.862 ± 0.002	0.375 ± 0.006
Gradient boosting	260	0.091	0.862	0.379	0.775	0.786	0.800	0.063	0.772	0.245	0.375	0.864 ± 0.002	0.381 ± 0.006

† *ROC-AUC* Receiver operating characteristic area under the curve, *AUPRC* Area under the precision-recall curve, *CV* Cross-validation

* Performance metrics include threshold-dependent measures (accuracy, balanced accuracy, sensitivity, specificity, precision, and F1 score) and threshold-independent measures (ROC-AUC and PR-AUC). Cross-validation results are reported for threshold-independent metrics to assess model stability

(Model A: balanced accuracy 0.760 ± 0.002, ROC AUC 0.832 ± 0.001, AUPRC 0.308 ± 0.003; Model B: balanced accuracy 0.737 ± 0.001, ROC AUC 0.805 ± 0.002, AUPRC 0.261 ± 0.004), indicating minimal variability across folds and confirming that the observed performance was not attributable to a favorable single train-test split.

As part of a sensitivity analysis to assess the impact of survey weighting, weighted logistic regression models were additionally fitted using BRFSS sampling weights. In the complete-case subset used for this analysis ($n = 40{,}073$), the weighted and unweighted prevalence estimates were 3.26% and 3.35%, respectively. The direction and relative magnitude of associations for key predictors were consistent between weighted and unweighted models (Supplementary Table 5), indicating that the main findings were not materially altered by the application of survey weights. Detailed year-specific odds ratios for all variables are provided in Supplementary Tables 6, 7, and 8.

Calibration analysis showed the following results. For Model A, the Brier score was 0.173, with a calibration intercept of -2.38 and a calibration slope of 1.00. For Model B, the Brier score was 0.187, with a calibration intercept of -2.37 and a calibration slope of 0.98. The slopes close to 1 indicate good agreement in relative probability estimation, whereas the negative intercepts suggest a tendency toward overestimation of absolute probabilities. Detailed calibration metrics are provided in Supplementary Tables 9, and calibration curves are shown in Supplementary Fig. 1.

Feature importance and model comparison

Feature importance analyses revealed substantial differences between Model A and Model B. Feature importance was quantified using impurity-based (Gini) importance derived from the final trained decision tree models, as described in the Methods section. The ranking and relative importance of the top predictors for each model specification are summarized in Supplementary Table 10.

In Model A, medication-related variables and measures of current health status ranked among the most influential predictors, consistent with their role as post-diagnosis indicators of disease management or severity. Notably, cholesterol-lowering medication use (cholmed3) accounted for the largest share of feature importance in Model A. In contrast, Model B exhibited a shift toward demographic, socioeconomic, and comorbidity-related variables. Employment status was retained as a secondary contributor in Model B, alongside stronger predictors such as age, hypertension, stroke history, and other chronic conditions. The relative importance of employment-related variables decreased compared with Model A but remained substantial, suggesting

Table 2 Predictive performance of decision tree models under inclusive and post-diagnosis-excluded specifications

Model specification	N_features	Threshold	ROC_AUC	AUPCR	Accuracy	Balanced accuracy	Sensitivity	Specificity	Brier score	Precision	F1 score	CV ROC-AUC (mean ± SD)	CV AUPRC (mean ± SD)
Model A (Inclusive)	333	0.50	0.832	0.309	0.709	0.760	0.822	0.699	0.173	0.202	0.324	0.832 ± 0.001	0.308 ± 0.003
Model B (post-diagnosis-excluded model)	321	0.50	0.806	0.260	0.678	0.738	0.809	0.666	0.187	0.183	0.299	0.805 ± 0.002	0.261 ± 0.004
Sensitivity analysis (excluding race)	318	0.50	0.806	0.260	0.678	0.738	0.809	0.666	0.187	0.183	0.299	Not evaluated	Not evaluated
Sensitivity analysis (excluding race and stroke history)	317	0.50	0.801	0.250	0.650	0.734	0.835	0.633	0.190	0.174	0.288	Not evaluated	Not evaluated

† *ROC-AUC* Receiver operating characteristic area under the curve, *AUPRC* Area under the precision-recall curve, *CV* Cross-validation

* Cross-validation was performed for the primary models (Model A and Model B) to evaluate model stability. Sensitivity models were conducted as prespecified analyses to assess the dependence of results on specific variable groups and were therefore not included in the cross-validation procedure

that these variables capture broader contextual correlates of prevalent CVD. Permutation-based importance analysis demonstrated broadly consistent rankings, with age, hypertension-related variables, stroke history, and chronic conditions remaining the most influential predictors. Employment status was retained as a contributing factor but was not among the dominant predictors. The ranking of the top 15 features is presented in Supplementary Table 11, while the full set of predictors is provided in Supplementary Table 12.

Discussion

In this population-based analysis of BRFSS data, our findings suggest that machine learning models can classify prevalent MI or CHD while explicitly addressing methodological concerns related to reverse causation and post-diagnosis information leakage. By implementing parallel model variants, we quantify the extent to which downstream disease consequences contribute to apparent predictive performance. Importantly, model performance remained robust even after removal of post-diagnosis proxy variables, and socioeconomic factors, including employment status, remained secondary predictors alongside established clinical risk factors in Model B.

Medication use, functional limitations, and perceived health status strongly influenced inclusive models, reflecting post-diagnosis treatment patterns and disease sequelae rather than upstream determinants. This pattern likely reflects reverse causality, as cholesterol-lowering medications are typically prescribed following diagnosis or clinical recognition of elevated cardiovascular risk. Therefore, the strong contribution of medication use in inclusive models should not be interpreted as evidence that the medication itself increases disease risk. Rather, it functions as a proxy indicator of prior clinical diagnosis and disease management. This distinction underscores the importance of separating inclusive classification models from screening models that aim to identify individuals prior to formal diagnosis. Importantly, exclusion of these variables resulted in only modest reductions in discriminative performance, indicating that meaningful signal remains beyond post-diagnosis information. Importantly, these models should not be interpreted as tools for prospective risk prediction, clinical screening, or real-world deployment, but rather as models for classifying prevalent disease status in a cross-sectional dataset.

The presence of employment status among the contributing variables in the Model B likely reflects the influence of multiple socioeconomic and physiological pathways associated with cardiovascular disease and should therefore be interpreted as a contextual correlate rather than a direct causal determinant. Employment status may reflect complex interactions among occupational

exposures, access to health care, functional capacity, and cumulative social disadvantage. In addition, employment status may capture physiological and psychological pathways linked to cardiovascular health. Chronic psychosocial stress related to unemployment, job insecurity, or high-demand occupations has been associated with sustained activation of the hypothalamic-pituitary-adrenal (HPA) axis and sympathetic nervous system, contributing to elevated cortisol levels, systemic inflammation, endothelial dysfunction, and adverse metabolic changes [14]. Employment instability may also be associated with depressive symptoms, sleep disturbances, and reduced engagement in preventive health behaviors, all of which are established contributors to CVD [15]. Thus, employment status in this cross-sectional context may serve as a marker of both social vulnerability and stress-related biological mechanisms.

Consistent with this interpretation, previous epidemiological studies have reported associations between unemployment and adverse cardiovascular outcomes [15–17]. Individuals who were unable to work or unemployed exhibited higher odds of CVD compared with those employed for wages. However, these estimates should be interpreted as descriptive associations and may reflect underlying health status and socioeconomic context rather than independent effects comparable to classic biomedical risk factors. These findings suggest that employment instability may contribute to cardiovascular risk through complex psychosocial and physiological pathways [18, 19].

The link between unemployment and CVD is supported by mechanistic evidence showing that chronic psychosocial and occupational stress is associated with heightened inflammatory activity, autonomic imbalance, and endothelial dysfunction, all of which contribute to cardiovascular pathophysiology [19–21]. These pathways suggest that work-related psychosocial stress may be associated with behavioral and biological processes relevant to cardiovascular health. Employment status therefore reflects not only economic stability but also psychosocial and lifestyle contexts that influence cardiovascular physiology. However, given the cross-sectional design of the present study, this association should be interpreted as a contextual correlate rather than evidence of a causal relationship. From an occupational health perspective, these findings highlight the relevance of work-related social conditions to public health understanding.

The identification of employment status as a contextual correlate carries implications for understanding population-level cardiovascular health patterns. These findings may inform population-level understanding of cardiovascular disease patterns, especially among individuals lacking traditional biomedical risk factors [8, 22]. These findings suggest that job stability, workplace conditions, and economic insecurity may be associated with population-level cardiovascular health patterns [15, 23]. Interpretable machine learning models may facilitate communication of these relationships by enabling transparent visualization of interactions among socioeconomic, behavioral, and clinical variables [24, 25]. The reported ORs represent descriptive associations derived from variable-specific (univariable) models and should not be interpreted as fully adjusted causal estimates.

This study has several limitations that should be considered when interpreting the findings. Because it relied on self-reported BRFSS data from a U.S. population, generalizability to other cultural or labor contexts may be limited. Similar concerns about the representativeness and recall bias of BRFSS data have been reported in prior methodological evaluations [26]. Replication in international cohorts with different employment structures would help validate these findings. Because the analysis is based on cross-sectional survey data, the models identify correlates of existing CVD rather than predictors of future cardiovascular events. Therefore, the results should not be interpreted as prospective risk prediction tools but rather as exploratory models for identifying population-level correlates of prevalent disease. Future research should incorporate longitudinal data to clarify the causal pathways linking employment instability, psychosocial stress, and cardiovascular outcomes. Integration of biomarker, occupational exposure, and workplace environmental data may further improve the predictive utility of such models [20, 27]. Although survey weights were not incorporated into the machine learning models, sensitivity analyses using weighted logistic regression yielded consistent results, supporting the robustness of the findings.

Conclusion

In summary, this study demonstrates that beyond conventional biomedical factors, occupational and socioeconomic conditions are associated with the presence of CVD in a large population-based survey. Machine learning models that incorporate these broader determinants can help characterize populations with a high burden of disease while accounting for potential post-diagnosis information leakage. The workplace represents not only a potential source of exposure but also an important context for understanding population-level cardiovascular health patterns. These findings highlight the value of interpretable AI approaches that integrate biomedical and social determinants in population-level cardiovascular health research.

Abbreviations

AI	Artificial intelligence
BRFSS	Behavioral Risk Factor Surveillance System
CHD	Coronary heart disease
COPD	Chronic obstructive pulmonary disease
CVD	Cardiovascular disease
HPA	Hypothalamic-pituitary-adrenal axis
MI	Myocardial infarction
OR	Odds ratio
PR-AUC	Precision–recall area under the curve
ROC	Receiver operating characteristic
U.S.	United States

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s12889-026-27570-3.

Supplementary Material 1.

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Authors' contributions
Jihyun Noh : Methodology, investigation, data curation, formal analysis, and writing – original draft preparation. Yonghwan Kim: Methodology, investigation, data curation, formal analysis, and writing – original draft preparation. Chaeseong Lim: data curation, formal analysis, visualization. Su-Yeon Cho: data curation, formal analysis, visualization. Yujin Kwon: data curation, formal analysis, visualization. Eun-Soo Lee: methodology, investigation, writing – reviewing, and editing. Won Kyu Kim: Conceptualization, visualization, project administration, funding acquisition, supervision, writing – reviewing, and editing. Yun Hak Kim : Conceptualization, visualization, project administration, funding acquisition, supervision, writing – reviewing, and editing. Kihun Kim : Conceptualization, visualization, project administration, writing – reviewing and editing, and supervision.

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Data availability
The datasets analyzed during the current study are publicly available from the BRFSS of the U.S. Centers for Disease Control and Prevention (CDC).

Declarations

Ethics approval and consent to participate
This study was a secondary analysis of publicly available, de-identified data obtained from the BRFSS, conducted by the U.S. Centers for Disease Control and Prevention (CDC). Informed consent was obtained from participants by the CDC at the time of the original BRFSS data collection. As the present study involved no direct interaction with participants and used only anonymized data, Institutional Review Board approval and additional informed consent were not required. The study was conducted in accordance with the principles of the Declaration of Helsinki.

Consent for publication
This manuscript does not contain any individual person's data in any form, including identifiable images or personal or clinical details.

Competing interests
The authors declare no competing interests.

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References

1. Virani SS, Alonso A, Aparicio HJ, Benjamin EJ, Bittencourt MS, Callaway CW, et al. Heart disease and stroke statistics—2021 update: a report from the American Heart Association. Circulation. 2021;143(8):e254–743.
2. Martin SS, Aday AW, Allen NB, Almarzooq ZI, Anderson CAM, Arora P et al. 2025 Heart Disease and Stroke Statistics: A Report of US and Global Data From the American Heart Association. Circulation. 2025, 151(8):e41-e660.
3. Oude Wolcherink MJ, Behr CM, Pouwels X, Doggen CJM, Koffijberg H. Health Economic Research Assessing the Value of Early Detection of Cardiovascular Disease: A Systematic Review. PharmacoEconomics. 2023;41(10):1183–203.
4. Powell-Wiley TM, Baumer Y, Baah FO, Baez AS, Farmer N, Mahlobo CT, et al. Social Determinants Cardiovasc Disease Circ Res. 2022;130(5):782–99.
5. Minhas AMK, Jain V, Li M, Ariss RW, Fudim M, Michos ED, et al. Family income and cardiovascular disease risk in American adults. Sci Rep. 2023;13(1):279.
6. Xia M, An J, Safford MM, Colantonio LD, Sims M, Reynolds K, et al. Cardiovascular Risk Associated With Social Determinants of Health at Individual and Area Levels. JAMA Netw Open. 2024;7(4):e248584.
7. Alpert JS. New Coronary Heart Disease Risk Factors. Am J Med. 2023;136(4):331–2.
8. Stringhini S, Carmeli C, Jokela M, Avendaño M, Muennig P, Guida F, et al. Socioeconomic status and the 25× 25 risk factors as determinants of premature mortality: a multicohort study and meta-analysis of 1· 7 million men and women. Lancet. 2017;389(10075):1229–37.
9. Krittanawong C, Virk HUH, Bangalore S, Wang Z, Johnson KW, Pinotti R, et al. Machine learning prediction in cardiovascular diseases: a meta-analysis. Sci Rep. 2020;10(1):16057.
10. Weng SF, Reps J, Kai J, Garibaldi JM, Qureshi N. Can machine-learning improve cardiovascular risk prediction using routine clinical data? PLoS ONE. 2017;12(4):e0174944.
11. Ambale-Venkatesh B, Yang X, Wu CO, Liu K, Hundley WG, McClelland R, et al. Cardiovascular event prediction by machine learning: the multi-ethnic study of atherosclerosis. Circul Res. 2017;121(9):1092–101.
12. Grimes DA, Schulz KF. Bias and causal associations in observational research. lancet. 2002;359(9302):248–52.
13. Kaufman S, Rosset S, Perlich C, Stitelman O. Leakage in data mining: Formulation, detection, and avoidance. ACM Transactions on Knowledge Discovery from Data (TKDD) 2012, 6(4):1–21.
14. Steptoe A, Kivimäki M. Stress and cardiovascular disease. Nat Reviews Cardiol. 2012;9(6):360–70.
15. Kivimäki M, Nyberg ST, Batty GD, Fransson EI, Heikkilä K, Alfredsson L, et al. Job strain as a risk factor for coronary heart disease: a collaborative meta-analysis of individual participant data. lancet. 2012;380(9852):1491–7.

16. Dupre ME, George LK, Liu G, Peterson ED. The cumulative effect of unemployment on risks for acute myocardial infarction. Arch Intern Med. 2012;172(22):1731–7.
17. Roelfs DJ, Shor E, Davidson KW, Schwartz JE. Losing life and livelihood: a systematic review and meta-analysis of unemployment and all-cause mortality. Soc Sci Med. 2011;72(6):840–54.
18. Matthews KA, Gallo LC. Psychological perspectives on pathways linking socioeconomic status and physical health. Annual Rev Psychol 2011, 62(1):501–30.
19. Steptoe A, Kivimäki M. Stress and cardiovascular disease: an update on current knowledge. Annu Rev Public Health. 2013;34(1):337–54.
20. Osborne MT, Shin LM, Mehta NN, Pitman RK, Fayad ZA, Tawakol A. Disentangling the links between psychosocial stress and cardiovascular disease. Circ Cardiovasc Imaging. 2020;13(8):e010931.
21. Ebong IA, Quesada O, Fonkoue IT, Mattina D, Sullivan S, Oliveira GMM, et al. The role of psychosocial stress on cardiovascular disease in women: JACC state-of-the-art review. J Am Coll Cardiol. 2024;84(3):298–314.
22. Marmot M, Allen J, Bell R, Bloomer E, Goldblatt P. WHO European review of social determinants of health and the health divide. lancet. 2012;380(9846):1011–29.
23. Berkman LF, Kawachi I, Glymour MM. Social epidemiology: Oxford University Press; 2014.
24. Ribeiro MT, Singh S, Guestrin C. Why should i trust you? Explaining the predictions of any classifier. In: *Proceedings of the 22nd ACM SIGKDD international conference on knowledge discovery and data mining*: 2016; 2016: 1135–1144.
25. Tjoa E, Guan C. A survey on explainable artificial intelligence (xai): Toward medical xai. IEEE Trans neural networks Learn Syst 2020, 32(11):4793–813.
26. Pierannunzi C, Hu SS, Balluz L. A systematic review of publications assessing reliability and validity of the Behavioral Risk Factor Surveillance System (BRFSS), 2004–2011. BMC Med Res Methodol. 2013;13(1):49.
27. Ayari H, Zeller M. Candidate Biomarkers to Evaluate the Association Between Psychosocial Stressors and Cardiovascular Diseases. A Short Review. Eur J Mental Health. 2023;18:1–12.

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